

Supply Planning Decision Report

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Business Case Demonstration | Nabu Casa | Supply Planning Decision Framework
Generated 2026-07-03

Executive Summary

Current WOS 4.8 On-hand weeks of supply	Planning Deficit 2,076 Units below target	Recommended New Buy 5,000 MOQ adjusted	Backlog Risk Critical Immediate escalation with supplier and executive review.
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The flagship product is currently classified as **Critical**. Inventory position is 18,000 units against a target of 20,076 units. The planning engine recommends a new buy of 5,000 units where the remaining deficit cannot be recovered through balancing, pull, or push actions.

Product And Supplier Position

Part Number	Description	Supplier	Supplier Status	Lead Time	Shipping
FLAGSHIP123	Smart Home Hub	CMHK	High Risk	8 weeks	2 weeks

Inventory Health

Target WOS 12 Planning policy	Current WOS 4.8 On hand only	Pipeline WOS 10.8 On hand plus open PO	Disruption WOS 7.2 On hand plus in transit
Target Inventory 20,076 Units	Inventory Position 18,000 Units	Open PO 10,000 Committed pipeline	In Transit 4,000 Already shipped

Purchase Order Health

PO	Quantity	Status	Ship Date	Delivery Date
PO10001	4,000	In Transit	2026-06-19	2026-07-03
PO10002	6,000	Open	2026-08-15	2026-08-29

Regional Demand And Allocation

Region	On Hand	Weekly Demand	Open Demand	Allocated	Backlog	Fill Rate
EU	6,000	693	3,000	3,000	0	100.0%
US	1,500	518	4,500	4,500	0	100.0%
APAC	500	462	2,000	500	1,500	25.0%

Optimization Sequence

Step	Planning Rule	Status	Recommendation
1	Stock Balance	FAILED	No Stock Balance Available
2	Oldest PO First	SUCCESS	Prioritise 1 Purchase Orders
3	Newest PO First	SUCCESS	Push 2,076 units to free OEM capacity
4	PO Shortage	REQUIRED	Planning Deficit Identified
5	Backlog Risk	Critical	Immediate escalation with supplier and executive review.

Planning Assumptions And Scope

Purchase Order Timeline

The 10,000-unit purchase order is assumed to have been created approximately 10 weeks before expected delivery: 8 weeks manufacturing lead time plus 2 weeks shipping and logistics.

Lead Time Definition

The stated 8-week lead time is treated as manufacturing lead time. Inventory planning uses total replenishment lead time because decisions depend on when stock becomes available for sale.

Regional Open Demand

Regional demand is treated as confirmed distributor and customer purchase orders awaiting fulfilment, not forecast demand.

Inventory Position

Open purchase orders already include in-transit inventory. Inventory position is calculated as on hand plus open purchase orders to avoid double-counting.

Distribution Network

A single central distribution centre is assumed where inventory is received before allocation to regional distributors.

Finished Goods Scope

The product is treated as a finished good supplied by a contract manufacturer. Service parts, repair inventory, reverse logistics, and RMAs are excluded.

Reusable Framework

The prototype is designed as a reusable supply planning decision framework supporting multiple products, suppliers, regions, and rule-based planning decisions.

Immediate Response Checklist

- Validate supplier root cause
- Confirm revised production schedule
- Review inventory health and weeks of supply
- Review customer backlog and open purchase orders
- Evaluate stock balance, pull, and push opportunities
- Calculate planning deficit and determine new buy requirement
- Communicate recovery plan

Planning Formula Reference

Formula	Definition
Target Inventory	Target WOS x Weekly Demand
Inventory Position	On Hand + Open Purchase Orders
Planning Deficit	Target Inventory - Inventory Position
Recommended New Buy	MAX(Planning Deficit, MOQ)